



Whichever Bin the Neighbours Already Rolled: A Street-by-Street Quasi-Experiment on Roster Age and the Shift from Reading to Copying on Bin Night

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Abstract. Kerbside waste-collection rosters that alternate a recycling week against a general week are common street furniture, usually a single laminated sheet fixed to a pole or letterbox, and the household-recycling literature has mostly asked whether a resident sorts correctly once a bin is already at the kerb rather than whether they consult a roster to decide which bin to present at all. We report a cross-sectional audit of 24 residential streets whose laminated rosters varied naturally in age from one to forty-one weeks since their last reprint, together with an embedded ablation in which eleven of the twenty-four streets received a QR-code freshness stamp linking to a live digital copy of the same schedule. Across a single collection-eve audit per street, the share of kerbed bins matching the roster's stated week fell as roster age increased ($r = -0.62$, $n = 24$ streets, $p < .001$), while the share matching only the nearest already-kerbed neighbour rose over the same range, from 4% on the freshest rosters to 20% on the stalest. The freshness stamp did not measurably narrow this gap. We interpret the pattern as a gradual handover from artefact-reading to social copying as a roster's authority quietly lapses, and discuss what it suggests about instructional signage that expects to be read anew each week.

Introduction

An alternating bin-night roster asks a household to do something a single sign rarely asks of a reader twice: consult it correctly on a different day each fortnight, indefinitely, without the roster itself changing in appearance to mark which week is current. Municipal recycling research has produced a substantial account of what happens once a bin reaches the kerb — which categories residents sort correctly, which items contaminate a load, and which interventions reduce sorting error — but has had comparatively little to say about the earlier decision the roster is meant to settle: which bin, of the two, goes out this week. That decision is treated in the literature as solved by the roster's presence, in the way a sign is treated as solved once it is installed.

Two adjacent literatures suggest it may not stay solved. Descriptive-norm research shows that

behaviour in view of others tracks the observable conduct of those others, sometimes more reliably than it tracks a posted instruction covering the same behaviour. Separately, work on habituation finds that a repeatedly encountered instructional artefact loses some of its power to redirect attention the longer it goes unchanged, even when its content has not itself gone out of date. Bin night sits at the intersection of both: it hands a household a printed instruction and, on the same morning, a street's worth of visible peer behaviour, and asks which one the household is actually using.

We report a cross-sectional, street-level audit that treats roster age as a naturally occurring quasi-experimental gradient rather than as something to manipulate directly. Three hedged contributions follow. First, we provide what we believe is the first estimate of how closely an alternating bin-night roster's stated schedule tracks household kerbing behaviour as a func-

tion of the roster's own printed age, across a range of ages arising naturally rather than staged as a single before/after reset. Second, we show that as roster-match declines, an alternative account of the same bin choice — matching the nearest already-kerbed neighbour rather than the roster — rises over the same range, consistent with a gradual handover between strategies rather than a simple loss of compliance. Third, we test, via an embedded ablation, whether a low-cost digital freshness cue attached to the existing roster narrows this gap; our results suggest it does not, at least not within one collection cycle.

Related work

Household recycling research finds that sorting errors — placing the wrong material in the right bin — are shaped by a household's uncertainty about category boundaries [1], set against a wider field of influences on recycling behaviour that a recent systematic review describes as broad and multi-level rather than reducible to any single lever [2]; this literature typically holds the choice of collection week fixed rather than treating it as its own outcome to explain. Behavioural interventions — reminders, pledges, and spillover from adjacent pro-environmental habits — can raise participation in an already-known kerbside schedule, with effects that vary by channel, dose, and mechanism [3], [4], [5].

A separate tradition treats the visible conduct of others as an instruction in its own right. Focus theory of normative conduct finds that whichever norm — descriptive or injunctive — is made locally salient shapes behaviour more than the other, and that a norm's grip depends on what a passer-by is actually attending to at the moment of choice [6], [7]. Deliberately engineered neighbour comparisons move household energy use and curbside recycling participation in the comparison's direction, an effect large enough to anchor a commercial industry [8], [9], [10]; manipulating the normative appeal displayed on a bin itself produces a comparable shift in what a passer-by treats as the default [11]. Economic models of informational cascades describe when it is individually rational to discount one's own information in favour of an immediately observable predecessor's choice, and field experiments confirm the effect holds outside the laboratory, in restaurant orders as much as financial markets [12], [13].

Instructional artefacts are known to lose some of their behavioural grip with repeated exposure even when unchanged, on a habituation timescale plausibly comparable to a roster's own reprint cycle, and to a household's own habit-formation timescale for the behaviour the roster is meant to cue [14], [15]. Whether a low-cost behavioural cue's own effect persists or decays is itself debated: several large-scale field experiments find that many nudges' apparent effects continue well after the intervention ends, an outcome attributed chiefly to durable changes in habit or technology rather than to the cue's continued visibility [16]. Slop University's own recent work has traced the same kind of gap between a stated instruction and enacted practice inside the household, and between a written artefact and an ad hoc channel that gradually substitutes for it [17], [18] — this paper asks the same question of a laminated sheet on a pole.

Method

We audited 24 residential streets within a single council's alternating-week collection zone, spanning three suburbs to avoid confounding roster age with any one neighbourhood's particular habits. Streets were selected to span as wide a range of roster ages as the zone's existing signage allowed, rather than to be representative of the zone as a whole. A street's roster age was read directly from a printed date stamp in the sheet's corner where present, and otherwise reconstructed from the account of a resident who could recall when the current sheet replaced its predecessor; the two methods agreed within a fortnight on every street where both were available.

On the eve of each street's next scheduled collection, two observers walked the street between 5:30 and 7:00am, recording for every household with a bin already at the kerb: which of the two bin types (general or recycling) had been rolled out; whether that type matched the roster's stated week for that street; and, separately, whether it matched the type already at the kerb of the nearest neighbouring household to have put a bin out before it (established from a fixed walking order down each side of the street). A bin was coded as copy-consistent only where it diverged from what the roster itself specified, so that a household correctly following the roster is never miscoded as merely copy-

ing a neighbour who happens to agree with it. Streets contributed a single audited collection eve each; the design is cross-sectional across streets rather than longitudinal within any one street.

Eleven of the twenty-four streets were selected, balanced across the resulting age tiers, to receive a small adhesive QR-code stamp affixed to the existing laminated roster four weeks before the audited collection eve, linking to a single-page, always-current digital version of the same schedule; the remaining thirteen streets' rosters were left as found. Recruitment and stamp placement proceeded street by street rather than household by household, since the roster itself is a shared artefact.

We summarise the roster-age relationship with a simple linear description of the match rate,

$$\text{match}_i = \beta_0 + \beta_1 \cdot \text{age}_i + \varepsilon_i,$$

fit across the 24 streets, and compare the freshness-stamp and no-stamp groups' match-rate distributions directly rather than assuming a parametric form, given the small number of streets in each arm.

Results

Figure 1 reports the roster-age relationship directly: streets audited within ten weeks of a reprint matched the roster's stated week on a tier mean of 93% of kerbed bins, falling to 85% for streets between eleven and twenty-five weeks past reprint, and to 76% beyond twenty-five weeks ($r = -0.62$, $n = 24$, $p < .001$). The relationship holds across all three suburbs; no single suburb's streets account for the trend on their own.

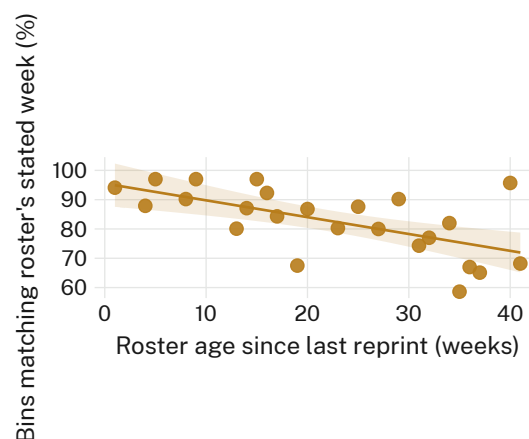


Figure 1: Roster-match rate falls as roster age rises across 24 streets ($r = -0.62$, $p < .001$); the freshest audited roster (1 week) matched 94% of kerbed bins, the oldest (41 weeks) 68%.

Figure 2 decomposes the same three age tiers into what a kerbed bin's type is actually explained by. Bins classified as copy-consistent — matching the nearest already-kerbed neighbour rather than the roster — rose from 4% of kerbed bins on the freshest rosters to 11% on ageing rosters and 20% on the stalest, while bins matching neither the roster nor a neighbour stayed roughly flat across all three tiers (3-4%). The rising share is copy-consistent rather than simply roster-inconsistent: as the roster's own match rate falls, the gap it leaves behind is filled disproportionately by agreement with a neighbour rather than by disagreement with everyone.

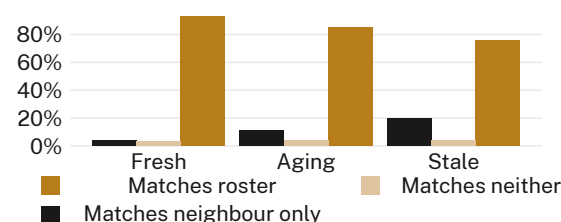


Figure 2: Copy-consistent bins (matching a neighbour, not the roster) rise from 4% to 20% of kerbed bins from the Fresh (≤ 10 wk) to the Stale (> 25 wk) tier, while bins matching neither stay flat.

Figure 3 reports the embedded ablation. The eleven streets given a QR-code freshness stamp four weeks before their audited collection eve showed a median roster-match rate of 80.3%, against 86.8% for the thirteen streets without one — a gap in the opposite direction to what the

stamp was intended to close, and one a small-sample comparison does not distinguish from noise at conventional thresholds ($p = .29$, n.s.). We retain the stamp in the reported protocol for completeness, on the view that a null result for a cheap intervention is worth reporting exactly because the intervention is cheap enough that a real effect, had one existed, would have been worth finding.

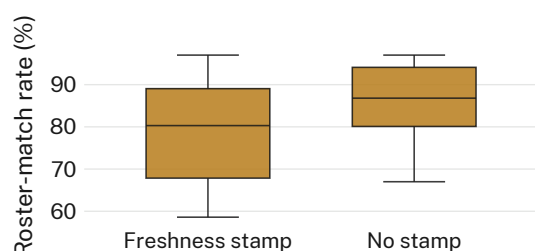


Figure 3: A QR-code freshness stamp does not raise roster-match rate (median 80.3% stamped vs 86.8% unstamped, n.s., $p = .29$).

Splitting the freshness-stamp comparison by age tier does not change the picture: within the ageing tier, stamped streets matched at a mean of 82.1% against 84.6% for unstamped streets; within the stale tier, 74.0% against 77.9%. Both differences run in the same, stamp-disadvantaged direction, and both are well within the range of between-street variation Figure 1 already shows at any single roster age.

Discussion

Read together, the three results suggest bin night runs on a slower and more mixed information diet than either “residents read the roster” or “residents copy their neighbours” would suggest on its own. Early in a roster’s life the two accounts are hard to tell apart, since neighbours are themselves mostly following the same roster; the value of tracking copy-consistent bins specifically, rather than roster-inconsistent bins generally, is that it lets the two accounts separate once the roster stops being the thing everyone is actually reading. Our results are consistent with a gradual handover in which social copying does not so much displace the roster as fill the space the roster’s own fading authority leaves behind, though we cannot rule out that both processes coexist from the roster’s first week and simply become visible in different proportions as it ages.

The freshness-stamp ablation suggests that the roster’s problem, on this evidence, is not legibility of its current content but something closer to a general erosion of attention that a fresher-looking version of the same artefact does not by itself restore. A more disruptive intervention — a full reprint, a differently coloured sheet, or a reminder delivered through a channel other than the roster itself — may behave differently, a question this design cannot answer.

Limitations

Our sample is limited to twenty-four streets within a single council’s alternating-week zone, though the consistency of the roster-age relationship across three distinct suburbs within that zone suggests the pattern is unlikely to be an artefact of any one street’s particular roster design. The cross-sectional design cannot rule out that streets with older rosters differ in some other way that itself predicts lower match rates; we found no relationship, however, between roster age and either street length or the number of households audited, the two most obvious candidate confounds available to us. A single audited collection eve per street cannot distinguish a stable pattern from a single unusually copy-heavy week, though the pattern’s consistency across three tiers of roster age argues against treating it as one week’s noise.

Conclusion

An alternating bin-night roster asks more of a household than a one-off sign does: correct use on a different day each fortnight, indefinitely, without the artefact itself changing to mark the shift. We find that the correspondence between what such a roster says and what a street actually does falls as the roster ages, that the gap left behind is filled disproportionately by copying an already-kerbed neighbour rather than by disagreement with everyone, and that a low-cost digital freshness cue does not close it. Future work might follow a single street’s roster continuously across a full reprint cycle rather than sampling many streets at once, and test whether a full reprint — rather than a cue layered onto the existing sheet — is what actually resets the pattern this audit finds decaying.

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